Automatic Multi-Class Brain Tumor Segmentation System Using Edge Aware Res-UNet

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in
Electronics Engineering

Submitted by

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DECLARATION

We hereby declare that the project report entitled **"Automatic Multi-Class Brain Tumor Segmentation System Using Edge-Aware Res-UNet"**, submitted in partial fulfillment of the requirements for the award of the **Bachelor of Technology (B.Tech) degree in Electronics Engineering** from **Dr. A.P.J. Abdul Kalam Technical University**, is an authentic record of original work carried out by us under the supervision and guidance of **Dr. Ashwini Kumar Upadhyay**, Assistant Professor, Department of Electronics Engineering, Rajkiya Engineering College, Kannauj.

We further declare that the content of this report has not been submitted, in whole or in part, for the award of any other degree or diploma at this or any other institution.

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The project report presents the results of independent research and study conducted by the students. The content of this report has not been submitted, in whole or in part, for the award of any other degree or diploma to the candidates or to any other individual.

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ABSTRACT

Brain tumor segmentation from multi-modal Magnetic Resonance Imaging (MRI) plays an important role in medical diagnosis, treatment planning, and tumor monitoring. Accurate identification of tumor regions such as necrotic core, edema, and enhancing tumor is a major challenge in medical image segmentation due to irregular tumor structures, unclear boundaries, and severe class imbalance in MRI data.

This project presents an automated multi-class brain tumor segmentation system using an Edge-Aware ResNet-UNet architecture. The work was carried out using the BraTS dataset containing four MRI modalities: T1, T1ce, T2, and FLAIR. Initially, a standard U-Net-based segmentation pipeline was implemented to establish a baseline model and validate the preprocessing and training workflow. Although the baseline model achieved satisfactory segmentation performance, it showed limitations in accurately capturing complex tumor boundaries and smaller tumor regions.

To improve segmentation quality, the project was further extended by developing an Edge-Aware ResNet-UNet model integrated with an edge attention mechanism. The proposed architecture uses a ResNet-based encoder for deep feature extraction [5] and incorporates boundary information generated using Canny edge detection [11]. The edge-aware mechanism helps the model focus more effectively on tumor boundaries during training, resulting in improved segmentation refinement and boundary preservation.

In addition to architectural improvements, a preprocessing strategy based on Active Image Slice Selection (AISS) and Adaptive Top-K Informative Slice Selection (ATISS) was implemented to reduce redundant MRI slices and minimize class imbalance. These preprocessing techniques improved training efficiency and enabled the model to learn more effectively from informative tumor-containing slices.

Experimental results demonstrate that the proposed edge-aware approach improves tumor segmentation performance, particularly in boundary-sensitive regions. The final trained model was also deployed using a Streamlit-based web application for interactive prediction and visualization. The proposed system provides an efficient and practical framework for automated brain tumor segmentation and can be further extended for advanced 3D medical imaging applications in future work.

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LIST OF ABBREVIATION

Abbreviation	Full Form
AI	Artificial Intelligence
AISS	Active Image Slice Selection
ATISS	Adaptive Top-K Informative Slice Selection
BraTS	Brain Tumor Segmentation
CE	Cross Entropy
CNN	Convolutional Neural Network
CSF	Cerebrospinal Fluid
DSC	Dice Similarity Coefficient
Ea+UNet	Edge-Aware ResNet-UNet
ED	Peritumoral Edema
ET	Enhancing Tumor
FCN	Fully Convolutional Network
FLAIR	Fluid Attenuated Inversion Recovery
GDL	Generalized Dice Loss
GN	Group Normalization
HGG	High Grade Glioma
IoU	Intersection over Union
LGG	Low Grade Glioma
MRI	Magnetic Resonance Imaging
NCR	Necrotic Core Region
NLP	Natural Language Processing
ReLU	Rectified Linear Unit
ResNet	Residual Neural Network
SVM	Support Vector Machine
T1ce	T1-weighted Contrast Enhanced
WT	Whole Tumor

CHAPTER 01

Introduction

This chapter introduces the fundamental concepts, motivation, and objectives behind the proposed brain tumor segmentation system. It discusses the importance of automated tumor segmentation in medical imaging and explains the role of multi-modal MRI in neuro-oncology. A brief overview of existing segmentation approaches is presented to highlight the evolution from traditional image processing methods to modern deep learning techniques. The chapter also outlines the major challenges associated with brain tumor segmentation, including irregular tumor boundaries and class imbalance, which motivated the development of the proposed approach. Finally, the objectives and key contributions of the project are presented to provide a foundation for the methodologies discussed in the following chapters.

1.1 Background and Motivation

Brain tumors are among the most lethal and debilitating forms of cancer, with high mortality rates worldwide. Accurate diagnosis, treatment planning, and patient monitoring heavily rely on precise visualization and delineation of the tumor regions. Magnetic Resonance Imaging (MRI) is the primary non-invasive imaging modality used in neuro-oncology because it provides exceptional soft-tissue contrast and can generate multiple structural sequences, such as T1-weighted (T1), T1-weighted with contrast enhancement (T1ce), T2-weighted (T2), and Fluid-Attenuated Inversion Recovery (FLAIR). Each of these modalities highlights different biological properties and sub-regions of a tumor, such as the enhancing core, the non-enhancing core, and the surrounding peritumoral edema.

Despite the wealth of information provided by multi-modal MRI, the manual analysis and segmentation of these scans by radiologists present significant challenges. Manual delineation is an extremely time-consuming and labor-intensive process, often requiring the analysis of hundreds of 3D volumetric slices per patient. Furthermore, it is highly subjective and prone to inter-rater and intra-rater variability, meaning different experts—or even the same expert at different times—may trace the boundaries of a tumor differently. Therefore, the development of robust, automated brain tumor segmentation algorithms is of paramount importance in modern **oncology**. Automated systems can provide rapid, reproducible, and objective tumor segmentations, significantly reducing the workload of clinical professionals and facilitating large-scale quantitative analyses for personalized medicine.

1.2 Previous Works and Related Studies

The field of automated brain tumor segmentation has evolved substantially over the past decade. Early approaches relied on classical image processing techniques such as thresholding, atlas-based methods, and hand-crafted feature extraction combined with traditional classifiers. While these methods provided an initial foundation, they were limited in their ability to generalize across the wide morphological variability observed in real-world brain tumor cases.

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), marked a transformative shift in the domain. Key developments include:

A. **CNN-based** segmentation architectures demonstrated the capacity to learn hierarchical feature representations directly from raw MRI data, outperforming traditional handcrafted approaches on benchmark datasets.

B. The **U-Net architecture**, introduced for biomedical image segmentation, became a landmark model due to its symmetric encoder-decoder structure and skip connections that preserve spatial information across resolutions [4]. It remains a foundational reference in the field.

C. Subsequent works **extended U-Net** with residual connections, attention gates, and dense connectivity to improve gradient flow and feature reuse during training [7], [12].

D. **The BraTS (Brain Tumor Segmentation) challenge**, organized annually, has been instrumental in benchmarking progress and providing standardized multi-modal MRI datasets for the community [1], [2], [3].

E. More recent efforts have explored 3D volumetric CNNs, **Transformer-based architectures**, and hybrid models to better capture long-range dependencies within volumetric MRI scans [14], [15], [17].

Collectively, these prior works establish the current research direction and identify persistent gaps—particularly in boundary precision and class imbalance handling—that motivate the present study.

1.3 Problem Statements & Objectives

While deep learning algorithms, particularly Convolutional Neural Networks (CNNs), have revolutionized medical image segmentation, several persistent challenges limit their clinical translation for brain tumor segmentation.

1.3.1 Challenges

- **Challenge 1: Fuzzy and Highly Irregular Boundaries**

  Brain tumors, especially high-grade gliomas such as glioblastomas, exhibit highly heterogeneous appearances, irregular shapes, and fuzzy, poorly defined boundaries. The tumors aggressively infiltrate surrounding healthy brain tissue, making it difficult to pinpoint exactly where the tumor ends and normal tissue begins.
  Traditional segmentation networks often produce less accurate or imprecise boundaries that fail to capture these intricate morphological details, which are critical for surgical planning and radiation therapy.

- **Challenge 2: Severe Class Imbalance**

  In a typical brain MRI volume, the actual tumor occupies only a very small fraction of the total cranial space, while the vast majority of the scan consists of healthy brain tissue and background. As a result, the model may learn background regions more effectively than tumor regions. Networks tend to over-predict the majority class (healthy tissue/background) to minimize overall error, leading to poor sensitivity in detecting small tumor regions or fine structural details.

1.3.2 Project Objectives

Develop an edge-aware ResNet-UNet model capable of improving tumor boundary detection using edge attention mechanisms.

1. **Standard U-Net:** To establish a robust baseline using the conventional U-Net architecture, which is widely regarded as the foundational model for biomedical image segmentation due to its symmetric encoder-decoder structure and spatial-preserving skip connections [4].

2. **Edge-Aware U-Net (Edge-Attention ResNet-UNet v2):** To design, implement, and evaluate a novel architecture that integrates a specialized parallel Edge Attention Branch. This branch extracts explicit boundary information **(lateral edge maps)** directly from the encoder features without modifying the main feature flow. These edge maps are then used to spatially modulate the features in the decoder via an attention mechanism, explicitly forcing the network to focus on refining fuzzy and irregular tumor boundaries.

3. **Optimize Data Preprocessing —** To develop and integrate a robust preprocessing pipeline using Active Image Slice Selection (AISS) to mitigate class imbalance and computational inefficiency [6]. AISS dynamically filters out non-informative 2D slices (containing only background or healthy tissue) from 3D MRI volumes, ensuring the model's capacity is focused on learning complex tumor features.

By selectively feeding the network only the most structurally informative slices, AISS optimizes the training process, accelerates convergence, and ensures the model's capacity is focused on learning complex tumor features rather than redundant background patterns.

1.4 Key Contributions of the Project

This project makes the following original contributions to the field of automated brain tumor segmentation:

1. **Edge-Aware Segmentation Model**

   A modified ResNet-UNet architecture with an edge attention mechanism was developed to improve tumor boundary detection and segmentation quality [4], [5], [7].

2. **AISS-Based Data Preprocessing**

   An Active Image Slice Selection (AISS) strategy was implemented to filter non-informative MRI slices and reduce class imbalance during training.

3. **Comparative Performance Evaluation**

The proposed model was compared with a standard U-Net baseline using evaluation metrics such as Dice Score, Sensitivity, and Specificity [1], [4].

4. **End-to-End Segmentation Pipeline**

   A complete pipeline including preprocessing, training, evaluation, and deployment was developed for automated brain tumor segmentation.

5. **Streamlit-Based Deployment**

   The final trained model was integrated into a Streamlit web application for interactive prediction and visualization.

The next chapter presents a detailed review of existing literature and recent advancements related to deep learning-based brain tumor segmentation.

CHAPTER 02

Literature Review

This chapter reviews the existing research and techniques related to medical image segmentation, with a primary focus on brain tumor segmentation using deep learning. It discusses the evolution of segmentation methods from traditional machine learning approaches to modern deep learning architectures such as CNNs and U-Net variants. The chapter also highlights important challenges including class imbalance, preprocessing complexity, and boundary preservation, which directly motivated the development of the proposed model used in this project.

2.1 Deep Learning in Medical Imaging

Medical image analysis has become an important application area of artificial intelligence and deep learning. For decades, the dominant approach to tasks such as lesion detection, organ delineation, and tumour segmentation was built on classical machine learning techniques — support vector machines (SVMs), random forest and active contour models among them. While these methods contributed meaningfully to early computer-aided diagnosis systems, they carried a fundamental limitation: every meaningful pattern had to be hand-crafted by domain experts before the algorithm could learn anything. Feature engineering of this kind was not only time-consuming but also inherently brittle; a feature designed for T1-weighted brain MRI, for instance, would rarely transfer to contrast-enhanced scans or images acquired on a different scanner. As a result, model generalisation was poor and real-world deployment remained largely aspirational.

2.1.1 The Shift to Convolutional Neural Networks

Traditional machine learning techniques used in medical image analysis mainly depended on manually designed features for tumor detection and segmentation. These methods required significant domain expertise and often failed to generalize well across different MRI datasets and imaging conditions.

The introduction of Convolutional Neural Networks (CNNs) significantly improved medical image segmentation by allowing models to automatically learn important image features directly from raw MRI scans. CNNs can effectively capture patterns such as edges, textures, shapes, and tumor structures through multiple convolution layers.

Later, Fully Convolutional Networks (FCNs) extended CNNs for pixel-wise image segmentation, where each pixel is assigned to a specific class. This advancement became highly important in medical imaging because accurate localization of tumor regions is essential for diagnosis and treatment planning. These developments eventually led to modern segmentation architectures such as U-Net and its variants.

2.1.2 The U-Net Architecture: A Landmark in Biomedical Segmentation

U-Net is one of the most widely used deep learning architectures for medical image segmentation. It was specifically designed for biomedical imaging tasks and became highly popular due to its ability to produce accurate segmentation results even with limited medical datasets [4].

The architecture follows an encoder-decoder structure. The encoder part extracts important features from the input image through convolution and pooling operations, while the decoder part reconstructs the segmentation map using upsampling operations. This structure helps the model learn both contextual and spatial information from MRI scans.

One of the key advantages of U-Net is the use of skip connections between the encoder and decoder layers. These connections transfer high-resolution spatial information directly to the decoder, helping the model preserve fine structural details during segmentation. This is especially useful in brain tumor segmentation, where accurate boundary localization is important [4].

Due to its strong performance and relatively simple architecture, U-Net has become a standard baseline model for many medical image segmentation tasks, including brain tumor segmentation using the BraTS dataset.

2.2 Overview of Existing Methods

With the growing availability of medical imaging datasets such as BraTS, deep learning-based methods have become the dominant approach for brain tumor segmentation. These models are capable of learning complex spatial and structural patterns from multi-modal MRI images, allowing accurate identification of tumor regions such as Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

Among different segmentation architectures, U-Net and its variants are widely used because they effectively combine contextual and spatial information through encoder-decoder structures and skip connections. Over time, researchers introduced several improvements to the original U-Net architecture to enhance segmentation accuracy and boundary detection.

Recent approaches include residual networks, attention mechanisms, transformer-based architectures, and lightweight segmentation models. Residual connections improve feature learning in deeper networks, while attention mechanisms help the model focus more on important tumor regions. Transformer-based models are capable of capturing long-range dependencies within MRI scans, whereas lightweight architectures aim to reduce computational complexity while maintaining good segmentation performance.

Although these approaches have achieved strong performance on benchmark datasets, challenges such as class imbalance, boundary preservation, and computational cost still remain important research problems. These limitations motivated the development of the proposed edge-aware segmentation approach used in this project.

2.2.1 U-Net and Its Variants

One of the most influential models in medical image segmentation is the U-Net architecture. It introduced a symmetric encoder–decoder structure with skip connections, allowing precise localization and better feature propagation.

Several studies have extended U-Net for brain tumor segmentation:

- A study using an **enhanced U-Net model (2022)** demonstrated strong performance on BraTS datasets, achieving Dice scores of approximately **0.95 for edema and 0.94 for enhancing tumor regions**, showing its effectiveness in multiclass segmentation tasks [7] .

- Another work proposed a **2D U-Net-based automated segmentation system (2023)** for BraTS2020, highlighting that U-Net can effectively distinguish tumor regions from healthy tissues while reducing manual effort [7], [20].

- Research also shows that optimizing U-Net parameters such as kernel size, activation functions, and dropout significantly improves segmentation accuracy, reaching up to **99.4% global accuracy** in some cases [18].

Despite its effectiveness, standard U-Net struggles with:

Class imbalance , Small tumor detection , Inclusion of irrelevant slices.

2.2.2 3D U-Net and Volumetric Models

To better capture spatial relationships across slices, researchers introduced 3D U-Net models, which process entire MRI volumes instead of individual slices.

- Studies show that 3D U-Net significantly improves segmentation of tumor subregions by learning inter-slice dependencies [15].
- However, these models require high computational power and memory, making them difficult to deploy in real-world scenarios.

Comparative research indicates that advanced frameworks like nnU-Net outperform earlier models by automatically adapting architecture and preprocessing strategies, achieving higher Dice scores across tumor regions [6].

2.2.3 Attention-Based Models

To address the issue of irrelevant features and improve focus on tumor regions, attention mechanisms were introduced.

- Attention U-Net enhances segmentation by selectively focusing on important regions while suppressing background noise [7].
- Recent research (2025) shows that attention-based U-Net architectures significantly improve segmentation accuracy and assist clinicians in better tumor localization [7], [8].
- Multi-scale attention models further enhance performance by capturing features at different resolutions, improving detection of small tumor regions [13], [14].

Although attention mechanisms improve accuracy, they increase model complexity and computational cost.

2.2.4 Hybrid and Transformer-Based Models

Recent advancements in medical image segmentation have increasingly focused on integrating transformer architectures with traditional convolutional neural networks (CNNs). While CNN-based models such as U-Net are highly effective at capturing local spatial features, they often struggle to model long-range dependencies and global contextual information within medical images. Transformer architectures, originally introduced in the field of Natural Language Processing (NLP), address this limitation through self-attention mechanisms that allow the model to capture relationships between distant regions of an image.

Recent advancements integrate transformer architectures with CNNs to capture long-range dependencies and global context.

- One notable example is **Res2Net-U-Net (2025)**, which enhances multi-scale feature extraction using Res2Net blocks integrated within the U-Net framework [4], [5]. This model demonstrated strong performance on brain tumor segmentation tasks, achieving Dice scores of **0.963 for Whole Tumor (WT), 0.951 for Tumor Core (TC), and 0.924 for Enhancing Tumor (ET)**. These results indicate a significant improvement over conventional U-Net-based models.

- Transformer-enhanced architectures such as **UNet++ variants** incorporate nested skip connections along with attention mechanisms to improve feature propagation and contextual learning [12]. The inclusion of global attention allows the network to better capture complex tumor patterns and spatial relationships, leading to more consistent segmentation outputs

- Another advanced architecture, **HART-UNet (Hybrid Attention Residual Transformer U-Net)**, integrates residual connections, attention modules, and transformer blocks into a unified framework [8], [17]. This model has achieved Dice scores of up to **0.97** on the BraTS dataset, highlighting its effectiveness in accurately segmenting brain tumors across multiple regions.

These models demonstrate state-of-the-art performance but come with significant drawbacks:

High computational requirements , Longer training time , Increased architectural complexity.

2.2.5 Lightweight and Efficient Models

To reduce computational cost, recent research focuses on lightweight architectures:

- **BTS U-Net (2025)** was proposed as a computationally efficient alternative, reducing training time by up to **79% while maintaining competitive performance** [18].
- These models aim to balance accuracy and efficiency, making them suitable for practical deployment.

However, lightweight models may compromise accuracy in detecting small tumor regions or complex structures.

2.2.6 Relevance to the Proposed Work

The limitations of existing segmentation methods, particularly in handling class imbalance, redundant training slices, and boundary preservation, motivated the development of the proposed approach in this project. The combination of AISS preprocessing and the edge-aware segmentation architecture was designed to improve segmentation efficiency while maintaining accurate tumor boundary detection.

2.3 Addressing Class Imbalance & Preprocessing

Class imbalance is one of the major challenges in brain tumor segmentation using MRI images. In the BraTS dataset, tumor regions occupy only a very small portion of the entire MRI volume, while most of the image contains healthy brain tissue and background. Due to this imbalance, segmentation models may become biased toward predicting background regions more frequently than tumor regions.

As a result, a model can achieve high overall accuracy while still failing to detect important tumor structures correctly. This problem directly affects tumor sensitivity and segmentation quality, especially for smaller tumor regions and unclear boundaries. Therefore, proper preprocessing and data balancing strategies are important for improving model performance and training stability.

2.3.1 Quantifying the Imbalance in 2D Slice-Based Training

The BraTS dataset contains multi-modal MRI volumes composed of multiple axial slices. However, only a limited number of slices contain visible tumor regions, while many slices contain only normal brain tissue or background information.

When all slices are used equally during training, the model receives a significantly larger amount of non-tumor data compared to tumor-containing slices. Due to this imbalance, the network may learn background patterns more effectively than tumor patterns. As a result, segmentation performance may decrease, especially for small tumor regions and irregular tumor boundaries.

This issue becomes more significant in 2D slice-based training because each slice is processed independently. Therefore, selecting informative slices and reducing unnecessary background data becomes an important step for improving model performance.

2.3.2 Common Techniques for Handling Class Imbalance

Several techniques have been proposed to reduce the effect of class imbalance in medical image segmentation tasks.

1. **Patch-Based Training**

   In patch-based training, smaller image regions are extracted from MRI scans instead of using the complete image. Tumor-centered patches are selected more frequently during training so that the model learns important tumor features more effectively. This method also reduces memory consumption during training.

2. **Data Augmentation**

   Data augmentation techniques artificially increase the diversity of training samples. Common augmentation methods include image rotation, flipping, scaling, intensity variation, and noise addition. These techniques help improve model generalization and reduce overfitting.

3. **Weighted Cross-Entropy Loss**

   Weighted Cross-Entropy Loss assigns higher importance to tumor classes during training. Since tumor pixels are fewer than background pixels, larger weights are given to tumor regions so that the model pays more attention to them during optimization.

4. **Dice Loss**

   Dice Loss is widely used in medical image segmentation because it directly measures the overlap between predicted segmentation and ground truth masks. It is more effective than standard loss functions when handling imbalanced datasets [9].

The Dice Similarity Coefficient (DSC) is defined as:

$$DSC = \frac{2\,|P \cup G|}{|P\,| + |G|} \qquad \ldots(2.1)$$

where:

- ***P** represents predicted tumor pixels*
- ***G** represents ground truth tumor pixels*

A higher Dice score indicates better segmentation performance.

5. Generalized Dice Loss (GDL)

Generalized Dice Loss is an extension of Dice Loss used for multi-class segmentation problems. It assigns balanced importance to different tumor sub-regions and helps improve segmentation performance for smaller and clinically important tumor areas [9].

2.3.3 Active Image Slice Selection (AISS)

Active Image Slice Selection (AISS) is a preprocessing strategy used in this project to reduce the effect of class imbalance in MRI data. In a complete MRI volume, many slices contain only background or healthy brain tissue without any visible tumor information. Training the model on such slices increases computational cost and provides limited learning benefit.

The main objective of AISS is to filter and retain only informative slices that contain meaningful tumor regions. By removing non-informative slices, the model receives more tumor-focused training data, which improves learning efficiency and segmentation performance.

In this project, AISS was implemented during the preprocessing stage before model training. Each MRI slice was analyzed using its corresponding segmentation mask. Slices containing sufficient tumor information were selected, while slices containing only background regions were excluded from the training dataset.

This approach helped reduce unnecessary computation, improved training convergence, and increased the model's focus on important tumor structures and boundaries.

2.4 Boundary and Edge-Aware Segmentation

Accurate boundary detection is one of the most important challenges in brain tumor segmentation. Although standard segmentation models such as U-Net can identify major tumor regions effectively, they often struggle to capture complex and irregular tumor boundaries. In many cases, the predicted segmentation masks become less accurate near tumor edges, especially in regions where tumor tissue gradually blends with healthy brain tissue.

To improve boundary localization, recent research has focused on edge-aware and boundary-guided segmentation techniques. These approaches provide additional structural information to the segmentation network so that it can better learn fine tumor details and boundary regions.

Several studies have introduced boundary-aware loss functions and edge-guided attention mechanisms to improve segmentation quality. Edge information helps the model focus more effectively on structural transitions between tumor and non-tumor regions. As a result, segmentation models become more sensitive to small tumor structures and irregular boundaries.

Inspired by these approaches, the proposed model in this project integrates an edge-aware attention mechanism within a ResNet-UNet architecture. A Canny edge detection method is used to generate boundary information from segmentation masks during training [11]. These edge maps guide the model to focus more on tumor boundaries and improve segmentation refinement.

The integration of edge-aware learning with deep feature extraction helps improve tumor localization, segmentation consistency, and boundary preservation in MRI images.

A. Multi-Task Learning for Edge Awareness : Another line of research focuses on **multi-task learning**, where segmentation and edge detection are learned simultaneously.

- **Oktay et al. (Attention U-Net, 2018)** indirectly improved boundary localization by guiding the network to focus on relevant regions, which enhances edge sensitivity [7].
- **Zhou et al. (UNet++, 2018)** introduced deeply supervised architectures that refine feature maps at multiple scales, leading to better boundary representation [12].
- More explicitly, **Zhang et al. (2019–2021)** proposed dual-branch networks where one branch performs semantic segmentation and the other detects edges, with shared encoders learning joint representations.

By jointly optimizing segmentation and edge detection tasks, these models encourage the encoder to learn features that are sensitive to both regional context and boundary details. This results in improved delineation of tumor subregions such as enhancing tumor and tumor core. However, these architectures increase training complexity and require careful balancing between tasks.

B. Attention Mechanisms and Edge Modulation: Recent works have explored integrating edge information directly into the feature extraction process using attention mechanisms.

- **Schlemper et al. (2019)** introduced attention-based gating mechanisms that highlight important spatial regions, indirectly improving boundary focus [8].
- **Qin et al. (2020, U²-Net)** incorporated nested U-structures that preserve fine spatial details, improving edge representation.
- Others generate edge maps internally using parallel convolutional branches and use them to modulate feature maps.

  In these approaches, edge maps act as **spatial attention guides**, emphasizing boundary regions during feature learning. By integrating edge information into intermediate layers, the network can better preserve high-frequency details that are typically lost during downsampling. This significantly improves the segmentation of fine tumor boundaries and transitions between different tumor classes.

This chapter reviewed the existing methods and research related to deep learning-based brain tumor segmentation. Different segmentation architectures, preprocessing strategies, and boundary-aware techniques were discussed along with the challenges associated with class imbalance and tumor boundary detection. The limitations of existing approaches motivated the development of the proposed edge-aware segmentation framework described in the following chapters.

CHAPTER 03

Dataset and Preprocessing Methodology

This chapter describes the dataset used in this study and the preprocessing techniques applied to prepare the data for effective model training. The research is conducted using the **Brain Tumor Segmentation (BraTS) dataset**, which provides multi-modal MRI scans along with expert-annotated ground truth labels for multiclass tumor segmentation. Each MRI volume consists of four modalities—T1, T1ce, T2, and FLAIR—capturing complementary anatomical and pathological information essential for identifying different tumor subregions.

However, raw MRI data presents several challenges, including intensity variations across scanners, presence of noise, class imbalance, and a large number of non-informative slices. These issues can significantly affect model performance if not properly handled. Therefore, a systematic preprocessing pipeline is essential to normalize the data, enhance relevant features, and ensure consistency across samples.

In this work, preprocessing includes modality-wise normalization, intensity standardization, and scaling to stabilize model training. Additionally, a data-centric strategy based on Active Image Slice Selection (AISS) and its enhanced version, Adaptive Top-K Informative Slice Selection (ATISS), is employed to filter out redundant slices and prioritize informative tumor-containing regions. This preprocessing framework plays a crucial role in improving training efficiency, reducing computational overhead, and enhancing the overall accuracy of multiclass brain tumor segmentation.

3.1 The BraTS Dataset

The accurate localization and segmentation of brain tumors are critical steps in clinical oncology for diagnosis, surgical planning, and survival prediction. The most commonly used dataset for developing and benchmarking automated algorithms in this domain is the Brain Tumor Segmentation (BraTS) challenge dataset. BraTS provides a large, multi-institutional repository of multi-modal magnetic resonance imaging (MRI) scans [1], [2], [3].

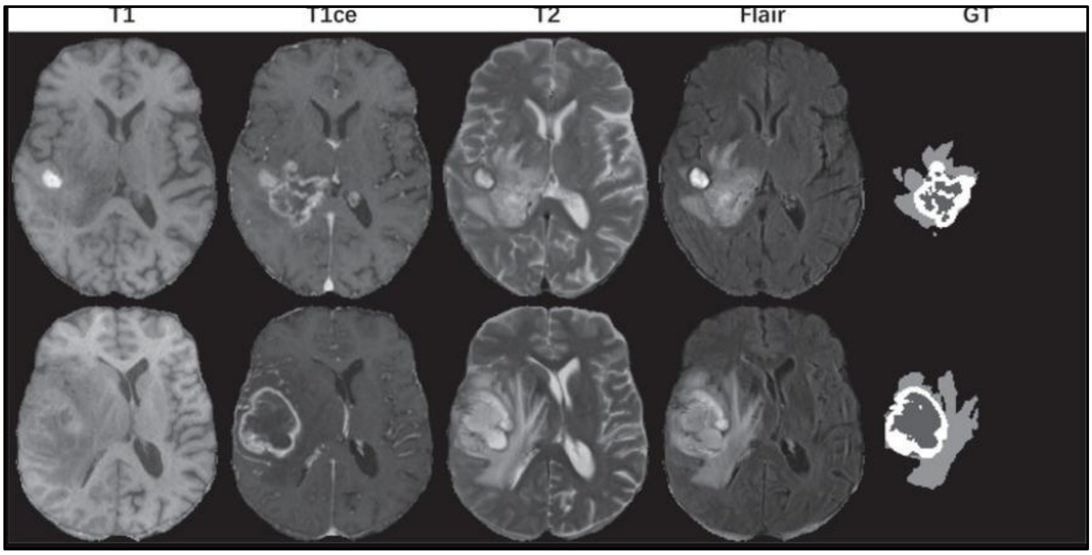

Figure 3.1. BraTS Dataset: MRI Modalities & GT Labels

Table 3.1. BraTS Dataset Summary

Property	Description
Dataset	BraTS
MRI Modalities	T1, T1ce, T2, FLAIR
Image Size	240×240×155
Classes	NCR, ED, ET
Format	.nii.gz
Segmentation Type	Multi-class

3.1.1 Understanding the MRI Modalities

MRI is the preferred imaging technique because it provides superior soft-tissue contrast without exposing the patient to ionizing radiation. The BraTS dataset typically includes four distinct structural MRI sequences for each patient, often referred to as multi-modal MRI. Each modality provides unique anatomical and pathological insights:

1. **T1-weighted (T1)** : **Basic Brain Anatomy**

   Provides excellent anatomical details of the brain, allowing for the clear visualization of healthy tissues. It is often used as a structural reference.

2. **T1-weighted with contrast enhancement (T1ce)** : **T1 + contrast agent**

   Blood-brain barrier broken by tumor - contrast leaks. ET glows bright

   Involves the injection of a gadolinium-based contrast agent. The agent accumulates in areas where the blood-brain barrier is disrupted, which is highly characteristic of active, malignant tumor growth. This modality is crucial for highlighting the active tumor core.

3. **T2-weighted (T2)** : **Full Tumor boundary**

   Shows fluid and swelling. Edema = very bright

   Highly sensitive to water content. It is excellent for delineating the overall extent of the tumor, including the fluid-filled peritumoral edema, though it may not distinguish well between different tumor sub-components.

4. **Fluid-Attenuated Inversion Recovery (FLAIR)**: Edema without fluid confusion

   Similar to T2 but specifically suppresses the signal from free-flowing cerebrospinal fluid (CSF). This makes FLAIR the most effective modality for distinguishing peritumoral edema from the surrounding healthy brain tissue and CSF.

   Edema - bright , CSF - dark

Raw Data

Each patient study includes four 3D volumetric MRI sequences: **T1, T1ce, T2, and FLAIR**. A standard BraTS 3D volume possesses the spatial dimensions of $240 \times 240 \times 155$, where 240×240 represents the height and width of a single axial slice, and 155 represents the total number of slices along the depth (z-axis) of the brain.

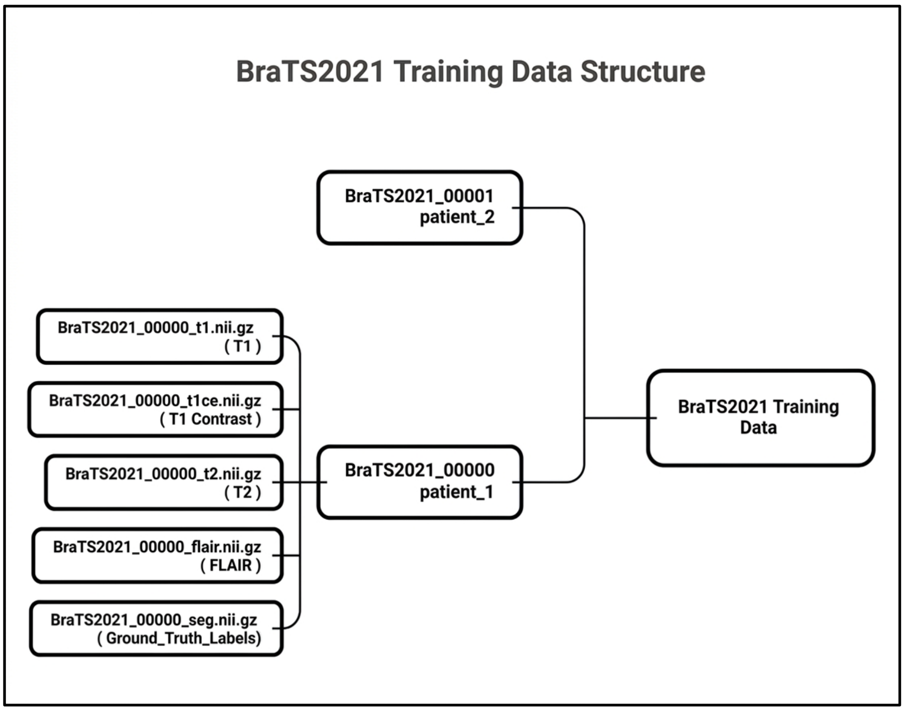

Figure 3.2. Data Structure

3.1.2 Tumor Sub-regions

The objective of the BraTS challenge extends beyond binary tumor localization to the precise classification of internal tumor heterogeneity into functionally and prognostically distinct sub-regions based on multi-parametric MRI signatures. Accurate delineation of these regions is critical for survival prediction and surgical planning.

1. **Necrotic Core (NCR)**:

   Represents necrotic (dead) tissue resulting from hypoxia when rapid tumor growth outpaces its blood supply — a hallmark of high-grade gliomas. It presents as hypo-intense (dark) regions within the tumor core on T1-weighted contrast-enhanced (T1ce) scans.The dead tissue area, typically found in the center of high-grade gliomas.

2. **Enhancing Tumor (ET)**:

   Represents the highly active, and rapidly proliferating tumor margins. It appears strongly hyper-intense (bright) on T1ce scans due to the microvascular breakdown of the blood-brain barrier (BBB), allowing gadolinium contrast extravasation. ET volume is a primary clinical indicator of malignancy grade. Appears hyper-intense (bright) on T1ce due to contrast agent leakage.

3. **Peritumoral Edema (ED)**:

   Refers to vasogenic swelling caused by fluid accumulation in the brain tissue adjacent to the tumor. While primarily fluid, this region frequently harbors infiltrative microscopic tumor cells undetectable by conventional imaging. It is clearly visualized as extended hyper-intense regions on Fluid-Attenuated Inversion Recovery (FLAIR) and T2-weighted scans.

For standardized clinical evaluation and model benchmarking, these sub-regions are aggregated into three overlapping anatomical metrics. The **Whole Tumor (WT)** is the union of all classes **(ED + ET + NCR)** and defines the complete volumetric extent of the disease including infiltrative boundaries. The **Tumor Core (TC)** is the union of **(ET + NCR)** , capturing the central bulk of the tumor that is typically targeted during surgical resection. The Enhancing Tumor (ET) is also evaluated on its own, as it reflects the most actively growing and malignant portion of the tumor. These three groupings allow segmentation models to be benchmarked not just on overall tumor detection, but also on their ability to distinguish clinically meaningful internal structures.

The research and experiments conducted in this project utilize the widely recognized Brain Tumor Segmentation (BraTS) challenge dataset. The BraTS dataset is universally considered the gold standard benchmark for evaluating state-of-the-art machine learning algorithms in neuro-oncology.

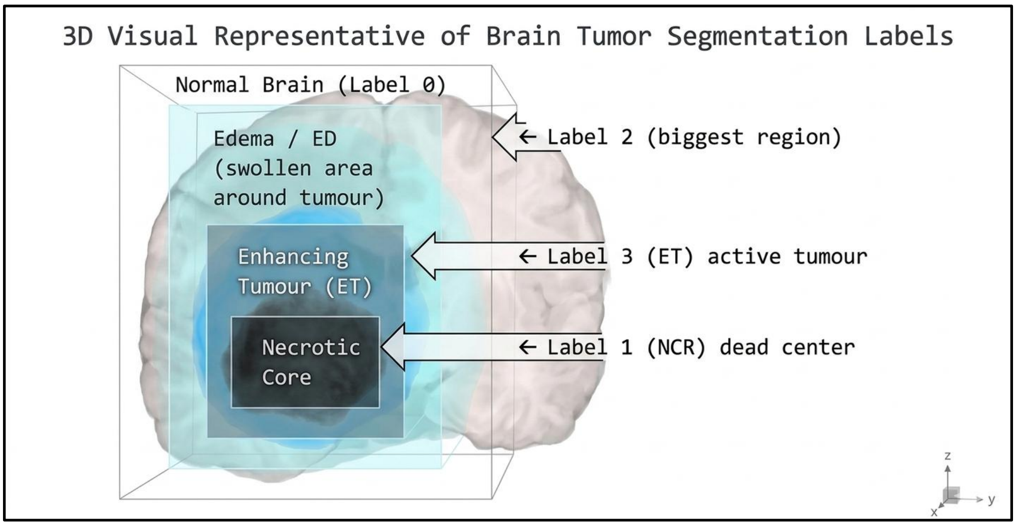

Figure 3.3. 3D Visual Representative of Brain Tumor Segmentations Labels

Annotations

Ground truth segmentation masks are provided for each patient volume, annotated by expert board-certified neuroradiologists. The annotations follow a specific multi-class labeling protocol:

→ Label 0: Background (and healthy brain tissue)

→ Label 1: Necrotic and Non-enhancing Tumor Core (NCR/NET)

→ Label 2: Peritumoral Edema (ED)

→ Label 4: GD-Enhancing Tumor (ET)

Data Source and Volume Dimensions

The dataset consists of multi-institutional pre-operative MRI scans from patients suffering from high-grade gliomas (HGG) and low-grade gliomas (LGG). The raw data is provided in the standard

Neuroimaging Informatics Technology Initiative (NIfTI) format (`.nii.gz`).

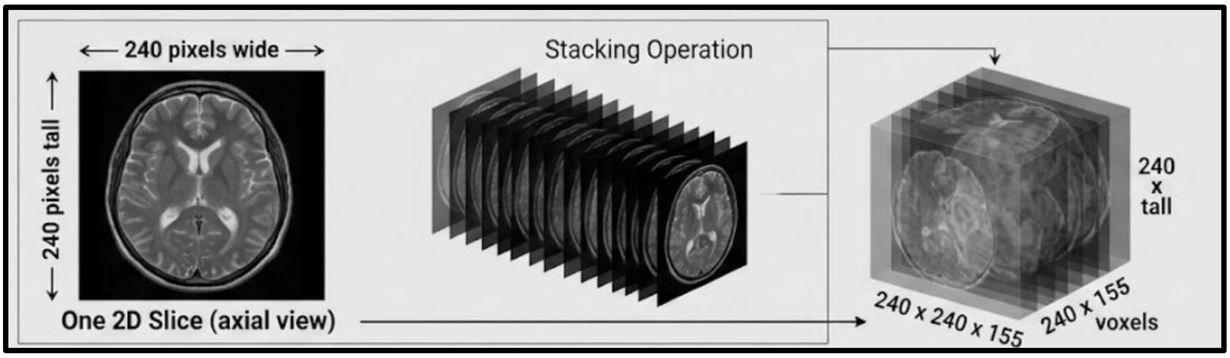

Figure 3.4. Representation of BraTS MRI Volume Dimensions (240×240×155)

Visual Representation of 1 slice

The four-panel visualization illustrates a multi-modal assessment of a single cerebral brain data slice from an example patient, demonstrating that each MRI channel provides critical, distinct information. The anatomical scan (CH 0) alone makes the tumor hard to detect, whereas the contrast-enhanced T1ce scan (CH 1) reveals a very obvious, bright, hyper-intense core, termed "ET glowing," that represents the active tumor. In contrast, the T2 scan (CH 2) visualizes a larger, diffuse area of edema surrounding the tumor, and the FLAIR scan (CH 3) suppresses cerebrospinal fluid (CSF) for a cleaner depiction of this edema, effectively differentiating the swelling from adjacent fluid spaces.

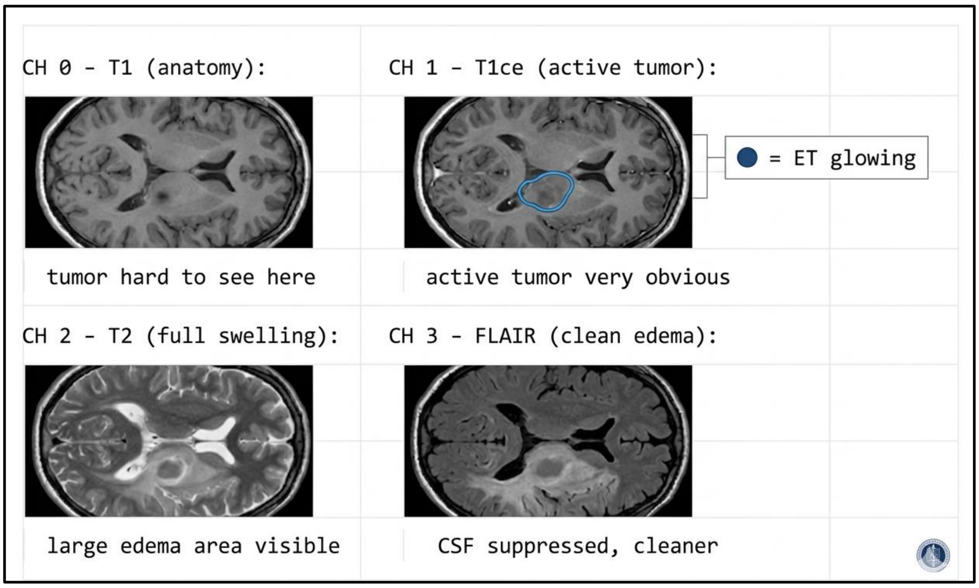

Figure 3.5. Multi-Modal Visualization of a Single MRI Slice Across T1, T1ce, T2, and FLAIR Modalities

3.2 Data Preprocessing Pipeline

MRI data inherently suffers from significant intensity variations across different scanners, institutions, and imaging protocols. Unlike natural images where pixel values represent absolute color (0-255), MRI voxel intensities have no fixed quantitative scale. Therefore, rigorous normalization is an absolute prerequisite before feeding the data into a deep learning model.

The preprocessing pipeline designed for this project involves the following systematic steps:

1. **Modality Independence:** Each of the four MRI modalities (T1, T1ce, T2, FLAIR) is normalized independently to preserve its unique signal characteristics
2. **Z-Score Normalization (Standardization):** To harmonize the intensity scales, Z-score normalization is applied. For each modality volume, the mean intensity of the brain region is subtracted from every voxel, and the result is divided by the standard deviation of the brain region. This ensures that the intensity distribution for every scan has a mean of zero and a unit variance

Crucially, background voxels (value 0) are excluded from the calculation of the mean and standard deviation to prevent them from skewing the statistical distribution of the actual brain tissue.

3. **Min-Max Scaling:** Following standardization, the voxel intensities are clipped to remove extreme outliers (e.g., top and bottom 1%) and subsequently scaled to a fixed range, typically `[0, 1]` or `[-1, 1]`. This stabilizes the gradients during the backpropagation phase of neural network training.

3.3 Active Image Slice Selection (AISS) vs. ATISS

To improve preprocessing efficiency, this project explored two slice selection strategies: Active Image Slice Selection (AISS) and Adaptive Top-K Informative Slice Selection (ATISS).

3.3.1 Active Image Slice Selection (AISS)

AISS removes MRI slices that contain only background or non-tumor regions. The purpose of this method is to reduce class imbalance and ensure that the segmentation model learns mainly from informative tumor-containing slices.

Although AISS improves training efficiency, some selected slices may still contain only small tumor regions with limited structural information.

3.3.2 Adaptive Top-K Informative Slice Selection (ATISS)

ATISS is an improved version of AISS proposed in this project. Instead of selecting all tumor-containing slices equally, ATISS prioritizes the most informative slices based on tumor visibility and structural content.

The method selects the top-K slices containing the highest tumor information from each MRI volume. This allows the model to focus more effectively on slices with clearer tumor structures and boundaries.

Compared to standard AISS, ATISS further reduces redundant training data and improves computational efficiency while preserving important tumor features required for segmentation.

3.3.3 Proposed ATISS: Adaptive Top-K Informative Slice Selection

Adaptive Top-K Informative Slice Selection (ATISS) is an enhanced slice selection strategy that dynamically identifies the most informative MRI slices based on tumour characteristics. Unlike fixed filtering methods, ATISS ranks slices using multiple criteria and selects the most relevant subset for training.

MRI Volume: **Shape**:(4, H ,W, D) 4 Modalities (T1, T1ce, T2, FLAIR)

Segmentation mask : (H, W, D) Multicast labels

Step-by-Step Algorithm

Step 1: Area-Based Filtering

For each slice index: $k \in [0, D-1]$

Extract Slice : $M_k = M[:,:,k]$ ...(3.1)

Compute tumour area: $area_k = \sum(M_k > 0)$ ...(3.2)

Compute Area ratio: $area\_ratio_k = \frac{area_k}{(H \times W)}$ ...(3.3)

Step 2: Filter Non_Informative Slices

If $area\_ratio_k < \gamma \Rightarrow$ discard slice

Step 3: Compute Slice Features

For each remaining slice :

1. **Tumour Area Score :** $area\_score_k = area\_ratio_k$ …(3.4)
2. **Class Diversity Score :** $classes_k = unique(M_k)$

   $num\_classes_k = | \{c \in classe_k \mid c > 0\} |$ ...(3.5)

   $class\_score_k = \frac{num\_classes_k}{max\_classes}$ ...(3.6)

3. **Boundary Complexity Score :** $edges_k = \nabla M_k$

   $boundary\_pixels_k = \sum(edges_k > 0)$ ...(3.7)

   $boundary\_scorek = \frac{boundary\_pixels_k}{area_k}$ ...(3.8)

Step 4: Compute Final Slice Score :

$$score_k = 0.5 \cdot area\_score_k + 0.3 \cdot class\_score_k + 0.2 \cdot boundary\_score_k \quad \textbf{...(3.9)}$$

This weighted combination ensures:

- Area importance (50%)
- Class diversity (30%)
- Boundary complexity (20%)

Step 5: Select Top-K Slices

$$\text{Sorted\_slices} = \text{sort\_by\_score\_descending(all\_valid\_slices)}$$
$$top\_k = sorted\_slices[0{:}K] \qquad \ldots(3.10)$$

Step 6 : Add Context Slices

$$final\_slices = \bigcup_{k \in top\_k} \{k-1, k, k+1\} \qquad \ldots(3.11)$$

$$\text{Subject to:} \qquad 0 \leq k + offset < D$$

Step 7: Ensure Rare Class preservation

If contains_rare_class (M[:,:,k]) ⇒ k ∈ final_slices

This ensures that **important but infrequent tumour types are not missed**.

Step 8: Remove Redundant Slices (Optional) :

$$\text{If similarity } (M_k, M_{k-1}) > 0.95 \quad \Rightarrow \quad remove\ k$$

CHAPTER 04

Model Architecture & Implementation Details

This chapter describes the deep learning models and implementation details used in this project for brain tumor segmentation. It first explains the baseline U-Net model developed to validate the preprocessing and training pipeline. The chapter then presents the proposed Edge-Aware ResNet-UNet architecture designed to improve tumor boundary segmentation. Finally, the training configuration, loss functions, and optimization settings used during model training are discussed.

4.1 Baseline U-Net Model

Before developing the proposed edge-aware segmentation model, a baseline U-Net architecture was implemented to validate the preprocessing pipeline and training workflow. The baseline model was developed using a ResNet-based encoder combined with a U-Net decoder structure for multi-class brain tumor segmentation.

4.1.1 Architecture Overview

The baseline segmentation model follows an encoder-decoder architecture based on the U-Net framework. A ResNetV2 backbone is used as the encoder to extract hierarchical feature representations from the multi-modal MRI input images.

- **Encoder**

  The encoder receives four MRI modalities (T1, T1ce, T2, and FLAIR) as input channels. Multiple convolutional and residual blocks are used to extract important spatial and structural features from the MRI scans. Max-pooling operations are applied to gradually reduce spatial dimensions and increase feature depth.

- **Decoder**

  The decoder reconstructs the segmentation map using upsampling operations and skip connections. Feature maps from the encoder are combined with decoder features to preserve important spatial information during reconstruction.

- **Segmentation Head**

  The final segmentation layer maps the extracted features into multiple tumor classes, including background, necrotic core, edema, and enhancing tumor regions.

The baseline model was used to establish the initial segmentation performance and evaluate the effectiveness of the preprocessing pipeline before introducing the proposed edge-aware architecture.

The baseline architecture (implemented as **ResNetUNet**) utilizes a classic symmetric encoder-decoder structure, but is supercharged with a pre-activation ResNet (ResNetV2) encoder.

4.1.2 Baseline Loss Functions and Clinical Limitations

To train the baseline segmentation model, a hybrid loss function combining Cross-Entropy Loss and Dice Loss was used. Cross-Entropy Loss helps improve pixel-wise classification accuracy, while Dice Loss improves the overlap between predicted segmentation masks and ground truth tumor regions.

The total loss function is defined as:

$$Loss_{total} = 0.5 \cdot CE(y, \hat{y}) + 0.5 \cdot Dice(y, \hat{y}) \quad \ldots \textbf{(4.1)}$$

The baseline model achieved satisfactory segmentation performance and successfully validated the preprocessing and training pipeline. However, during evaluation, the model showed limitations in accurately segmenting complex tumor boundaries and smaller tumor regions.

Since the standard segmentation network mainly focuses on region-based learning, some fine structural details near tumor boundaries were not captured effectively. This limitation motivated the development of the proposed edge-aware segmentation architecture described in the next section.

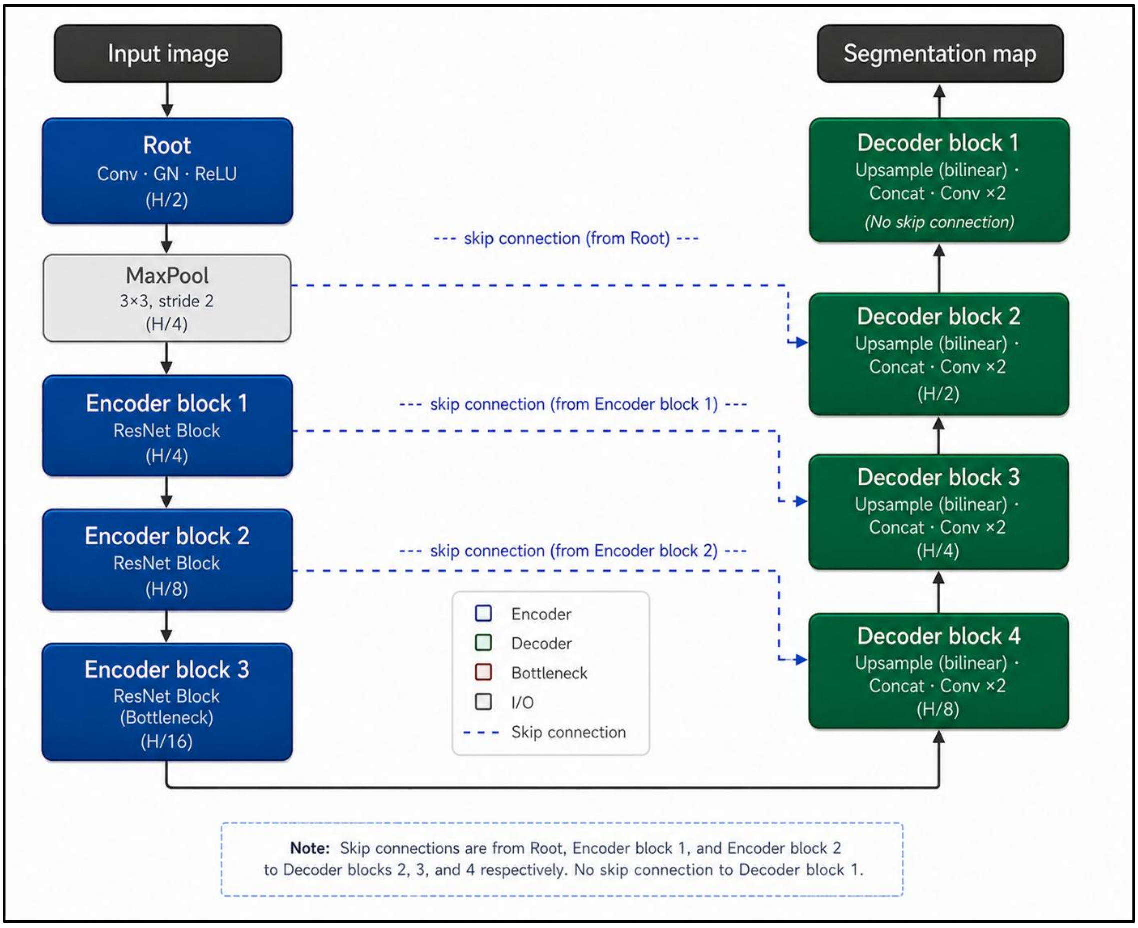

Figure 4.1. ResNet-based U-Net architecture for multi-scale image segmentation.

4.2 Proposed Edge-Aware ResNet-UNet Architecture

The limitations observed in the baseline segmentation model motivated the development of the proposed Edge-Aware ResNet-UNet architecture. The main objective of this model is to improve tumor boundary segmentation by integrating edge-aware learning with deep feature extraction.

The proposed architecture combines a ResNet-based encoder, a U-Net decoder, and an additional edge attention branch. The edge-aware mechanism helps the network focus more effectively on tumor boundary regions during training, improving segmentation refinement and structural consistency.

4.2.1 ResNet Encoder and Deep Feature Extraction

The proposed model uses a ResNetV2-based encoder for extracting deep spatial and structural features from multi-modal MRI images. The encoder processes four input modalities (T1, T1ce, T2, and FLAIR) simultaneously and generates hierarchical feature representations required for accurate tumor segmentation.

Residual blocks are used within the encoder to improve feature learning in deeper layers of the network. These blocks help preserve important information during training and improve the overall learning capability of the model.

The encoder gradually reduces the spatial dimensions of the input image through downsampling operations while increasing feature depth. Intermediate feature maps generated at different encoder stages are stored and later connected to the decoder through skip connections.

These extracted feature representations provide important contextual and structural information for accurate segmentation of tumor regions and boundaries.

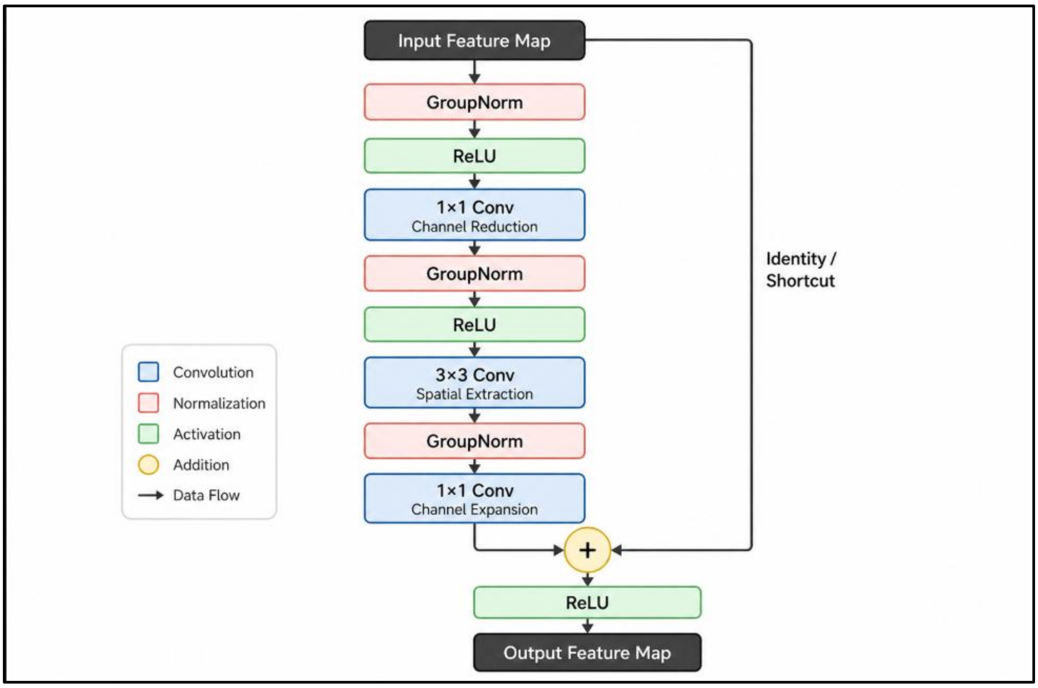

Figure 4.2. ResNetV2 Residual Block Structure

4.2.2 The Edge Attention Branch: Ground Truth Boundary Generation

The proposed model includes an additional edge attention branch to improve tumor boundary segmentation. The purpose of this branch is to provide boundary-focused information during training so that the model can better identify irregular and unclear tumor edges in MRI images.

The edge attention branch receives intermediate feature maps from the encoder and processes them through convolution layers to generate edge-focused feature representations. These features are used to guide the decoder toward more accurate boundary prediction during segmentation.

To generate boundary information, Canny edge detection is applied to the ground truth segmentation masks during preprocessing. The generated edge maps act as additional supervision signals for the edge attention branch during training.

The predicted edge maps are further integrated with decoder features using an attention mechanism. This helps the segmentation network focus more effectively on important tumor boundary regions and improves segmentation refinement near complex tumor structures.

The integration of edge-aware learning improves boundary preservation and helps produce more consistent segmentation outputs for different tumor regions.

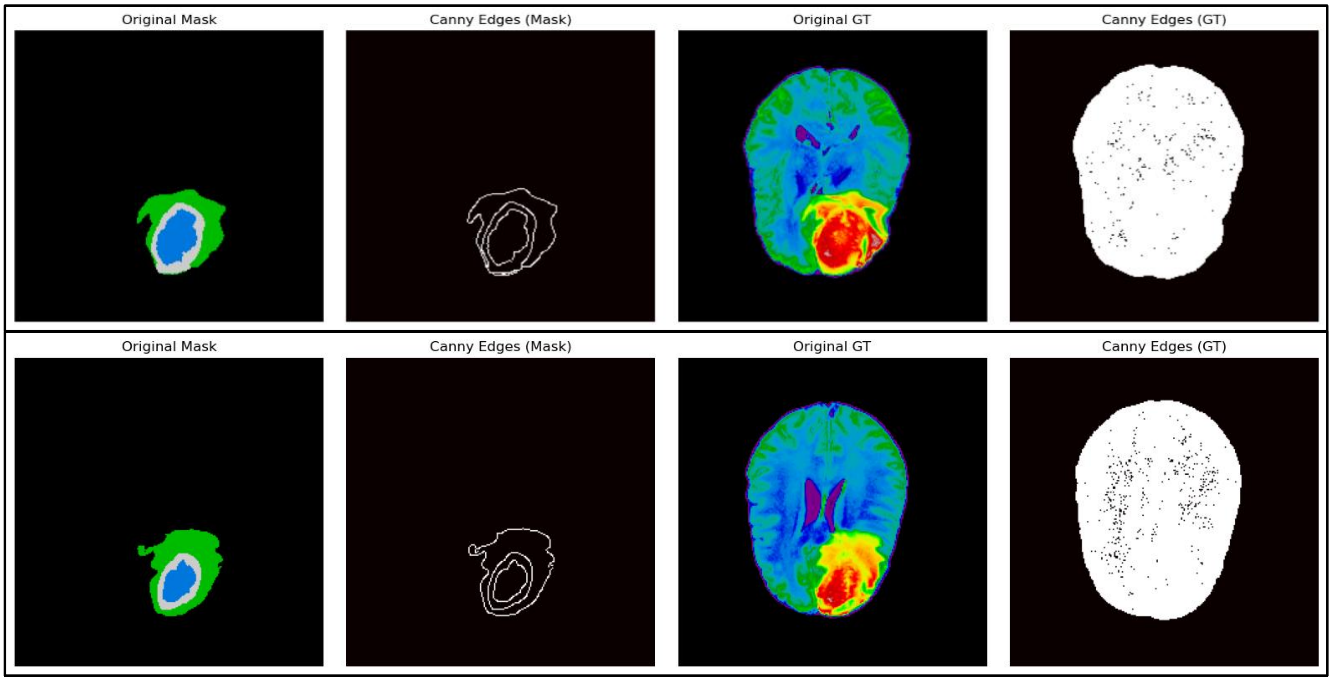

Figure 4.3. Ground Truth Mask and Generated Edge Map

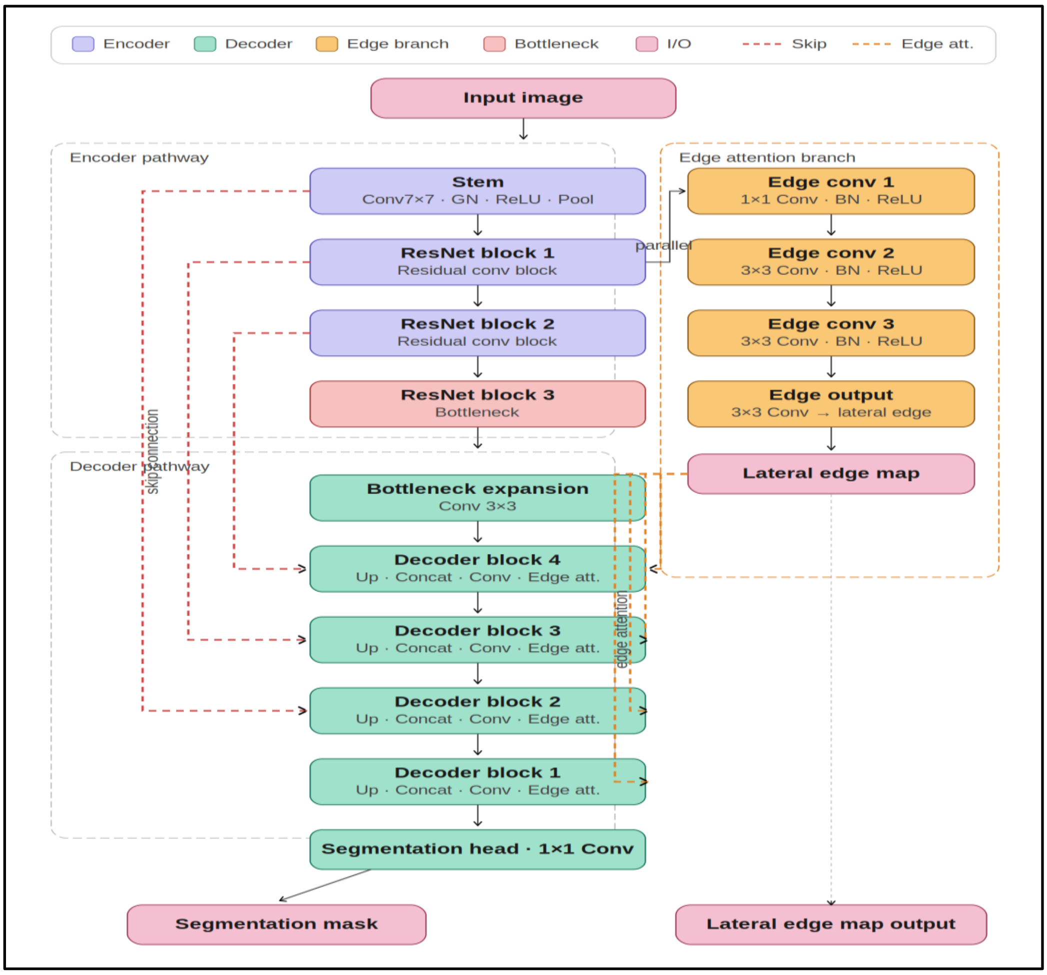

Figure 4.4. Proposed Edge-Aware ResNet-UNet Architecture

4.2.3 Specialized Boundary-Aware Hybrid Loss Function

To simultaneously train the main segmentation head and the parallel edge branch, we engineered a complex tripartite hybrid loss function. The total loss for each batch is calculated as:

$$Loss_{total} = 0.4 \cdot loss_{CE} + 0.6 \cdot loss_{Dice} + 0.5 \cdot loss_{Edge} \qquad \ldots(4.2)$$

- **Weighted Cross-Entropy ($loss_{CE}$):** Evaluates pixel-level classification on the main decoder output. To combat the severe class imbalance (where healthy tissue dominates), we applied static class weights: Background is heavily down-weighted (0.1), while the critical Necrotic Core (NCR/NET) and Enhancing Tumor (ET) classes are up-weighted (2.0 and 1.5, respectively).

- **Generalized Multiclass Dice Loss ($loss_{Dice}$):** Evaluates the region-based overlap (Intersection over Union). We utilized a Generalized version that dynamically calculates class weights inversely proportional to the squared volume of the class in the current batch. This forces the network to heavily prioritize tiny, hard-to-detect tumor sub-regions that standard Dice loss would ignore.

- **Boundary Loss ($loss_{Edge}$):** This loss exclusively supervises the lateral_edge output of the

Edge Attention Branch. It combines standard Binary Cross-Entropy (BCE) against the Canny-generated 1-pixel edges with a Distance Transform Regularization. Using a Signed Distance Function (SDF) map, the loss heavily penalizes edge predictions that fall far away from the true boundary, while remaining forgiving of microscopic offsets near the fuzzy tumor margins.

4.3 Training Configuration & Hyperparameters

The proposed segmentation models were trained using multi-modal MRI images from the BraTS dataset. Training was performed using GPU acceleration to improve computational efficiency and reduce training time.

The AdamW optimizer was used for parameter optimization because of its stable convergence performance in deep learning applications. A fixed learning rate strategy was initially used and later adjusted during training to improve model convergence.

To improve training stability and reduce overfitting, preprocessing techniques such as normalization, slice selection, and data filtering were applied before training. The proposed ATISS preprocessing strategy helped reduce redundant MRI slices and improved training efficiency.

The segmentation models were trained using mini-batch learning for multiple epochs until stable validation performance was achieved. Evaluation was performed using Dice Score, Sensitivity, and Intersection over Union (IoU) metrics.

Table 4.1. Training Configuration

Parameter	Value
Optimizer	AdamW
Learning Rate	0.01
Batch Size	8
Epochs	50
Input Size	192×192
Loss Function	Hybrid Loss
Dataset	BraTS

Hardware Optimization & Memory Management To optimize tensor operations and accelerate the training process without sacrificing model depth, we employed the following PyTorch-native optimizations:

- **Automatic Mixed Precision (AMP):** Model weights and gradients were dynamically cast to float16 during the forward pass. This significantly reduced overall memory consumption and accelerated training iterations by approximately 30% without degrading segmentation accuracy.

- **TensorFloat-32 (TF32):** We enabled native TF32 support to optimize matrix multiplication speed while maintaining high numerical precision.

- **Channels Last Memory Format:** The model architecture and input tensors were cast to memory_format=torch.channels_last, heavily optimizing the speed of the 2D convolutions on the GPU.

- **Resolution Scaling:** The standard 256×256 BraTS slices were resized to 192×192, establishing an optimal balance between preserving fine-grained spatial features and maintaining computational efficiency.

4.3.1 Data Configuration and ATISS

- **Dataset Subset:** The training pipeline utilized a representative subset of 500 cases from the full BraTS dataset, uniformly split into 90% for training and 10% for validation.

- **Adaptive Slice Selection:** Instead of training on all 155 slices per patient volume—many of which contain only empty background—we utilized our Adaptive Top-K Informative Slice Selection (ATISS) algorithm. We set k=10, meaning the pipeline autonomously selected the 10 most tumor-dense slices from each of the 450 training cases. This yielded a highly concentrated, informative dataset of exactly 4,500 training slices.

- **Batching:** Based on the resolution and AMP optimizations, the batch size was set to 8, providing stable gradient updates.

4.3.2 Optimization Algorithm and Learning Rate Strategy

- **Optimizer:** We utilized Stochastic Gradient Descent (SGD) as the primary optimizer. It was configured with a base learning rate of 0.01, a momentum factor of 0.9, and a weight decay to prevent overfitting.

- **Scheduler:** To ensure smooth convergence and prevent the network from getting stuck in local minima, we applied a Cosine Annealing Learning Rate Scheduler. This progressively decayed the learning rate following a cosine curve over the course of training.

- **Epochs & Convergence:** The model was trained for a maximum of 50 epochs. Given the concentrated nature of the ATISS-filtered data and the highly efficient pre-activation ResNet backbone, 50 epochs proved entirely sufficient to observe clear loss convergence and stabilization of the validation metrics.

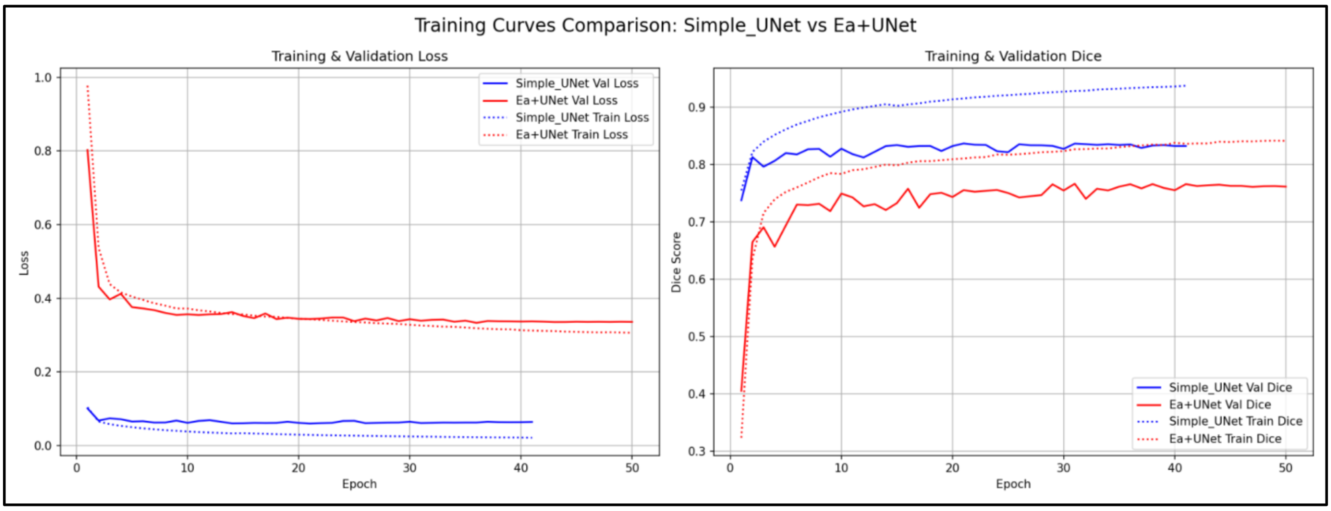

Figure 4.5. Training Curves Comparison of U-Net + EA architecture

CHAPTER 05

Results and Evaluation

To comprehensively assess the performance of both the baseline pipeline and the proposed Edge-Aware architecture, we employed a rigorous evaluation strategy. The models were evaluated not just on raw volumetric overlap, but also on their clinical reliability and their ability to accurately delineate complex morphological boundaries.

5.1 Evaluation Metrics:

The segmentation performance was quantified using three distinct categories of metrics: overlap metrics, clinical diagnostic metrics, and boundary distance metrics.

5.1.1 Overlap Metrics (Dice IOU):

Overlap metrics evaluate how much the predicted segmentation mask spatially aligns with the ground truth masks, quantifying the direct intersection between the model's prediction and the ground truth mask.

- **Dice Similarity Coefficient (DSC):** The Dice score is the primary metric used in medical image segmentation. It measures the harmonic mean of precision and recall, penalizing both false positives and false negatives. A Dice score of 1 indicates perfect overlap, while 0 indicates no overlap.

$$DSC = (2 * \frac{|X \cap Y|}{|X| + |Y|}) \qquad \ldots(5.1)$$

where X is the prediction and Y is the ground truth

- **Intersection over Union (IoU / Jaccard Index):** Similar to the Dice score, IoU measures the overlapping area divided by the total area of union between the predicted and ground truth masks. It tends to penalize single instances of bad classification more heavily than the Dice score.

$$IoU = \frac{|X \cap Y|}{|X \cup Y|} \qquad \ldots(5.2)$$

5.1.2 Clinical Metrics:

While overlap metrics are excellent for overall volume agreement, clinical metrics are essential for understanding how well the model detects the actual presence of pathological tissue (like the necrotic core or enhancing tumor), minimizing missed diagnoses.

- **Sensitivity (Recall / True Positive Rate):** Sensitivity measures the proportion of actual positive pixels (tumor tissue) that were correctly identified by the model. In a clinical setting, high sensitivity is crucial, as missing a portion of the tumor (false negatives) can be fatal. This metric heavily influenced our evaluation of the Ea+UNet.

$$Sensitivity = \frac{TP}{TP + FN} \quad ...(5.3)$$

(TP = True Positives, FN = False Negatives)

- **Specificity (True Negative Rate):** Specificity measures the proportion of actual negative pixels (healthy brain tissue/background) that were correctly identified. Given that the background constitutes the vast majority of an MRI scan, specificity usually remains very high, but it is tracked to ensure the model isn't simply over-predicting tumor tissue everywhere.

$$Specificity = \frac{TN}{TN + FP} \quad ...(5.4)$$

(where TN = True Negatives, FP = False Positives)

5.1.3 Boundary Metrics

Because standard overlap metrics can be insensitive to complex, jagged tumor boundaries, we utilize spatial distance metrics to evaluate how well the predicted edges align with the ground truth edges.

- **Hausdorff Distance (HD95):** The Hausdorff Distance measures the maximum distance between the predicted boundary and the ground truth boundary. Because the absolute maximum distance is highly sensitive to single-pixel outliers, we use the 95th percentile Hausdorff Distance (HD95). A lower HD95 value indicates that the boundaries of the predicted segmentation closely follow the true anatomical boundaries.

- **Boundary F1 Score (BF Score):** The Boundary F1 Score computes the precision and recall specifically along the contour of the segmented object within a small distance threshold. It is an excellent measure of how accurately the model delineated the crisp edges of the tumor, which is the primary objective of our Ea+UNet architecture.

5.2 Quantitative Results

The models were evaluated against the validation subset of the BraTS dataset. The results are analyzed in the context of our developmental journey: from establishing a functional baseline to engineering a clinically safer solution.

5.2.1 Baseline Pipeline Validation

The standard U-Net approach served as our initial baseline to validate the data pipeline and preprocessing methodologies. This model yielded strong overall overlap scores, achieving a Dice Score of **0.7355** and an IoU of **0.6455**. Furthermore, it demonstrated exceptional precision in avoiding healthy tissue, achieving a near-perfect Specificity of **0.9988**.

However, a deeper analysis of the clinical metrics revealed a significant vulnerability: the model's

Sensitivity was constrained to **0.7563 (75.63%).** This indicated that the standard convolutional approach was overly conservative. It struggled to capture the diffuse, ambiguous margins of the Glioblastomas, resulting in the model frequently missing true malignant tissue (false negatives).

Table 5.1 Baseline Model Evaluation Metrics

Class	Dice Score	IoU	Sensitivity	Specificity
NCR/NET	0.680	0.603	0.678	0.999
ED	0.728	0.620	0.738	0.998
ET	0.797	0.712	0.851	0.998
Foreground Mean	**0.735**	**0.645**	**0.756**	**0.998**

The baseline ResNet-UNet model achieved satisfactory multiclass segmentation performance across the major tumor regions. The model produced the highest Dice score for the Enhancing Tumor (ET) region with a Dice coefficient of 0.797, indicating effective detection of active tumor regions. However, comparatively lower Dice scores for NCR/NET and edema regions revealed limitations in handling irregular boundaries and small tumor structures. These observations motivated the development of the proposed edge-aware architecture.

In the context of medical imaging, Sensitivity is arguably the most critical metric. A higher Sensitivity indicates that the Ea+UNet is significantly better at detecting the actual presence of pathological tissue (reducing False Negatives). The Simple UNet's higher Dice score alongside lower Sensitivity suggests that it behaves very conservatively—only predicting pixels it is highly confident about, thereby maximizing its overlap ratio but missing harder-to-detect tumor regions. The Ea+UNet, guided by the structural edge map, takes a more clinically viable approach by ensuring more of the total tumor mass is identified.

Furthermore, the Ea+UNet achieved an excellent Hausdorff Distance (HD95) of 5.06 mm, demonstrating that its predicted contours closely hug the true anatomical boundaries.

5.2.2 Proposed Model Performance (Ea+UNet)

The proposed Edge-Aware ResNet-UNet v2 was engineered specifically to address the structural shortcomings of the baseline. When evaluated, the Ea+UNet demonstrated highly competitive overlap, with a Dice Score of **0.7181** and an IoU of **0.6187.** Crucially, the model maintained an exceptional Specificity of **0.9984**, proving that the addition of the Edge Attention Branch did not cause the model to hallucinate tumors in healthy tissue.

Furthermore, the Ea+UNet established strong morphological alignment, achieving an HD95 score of **5.0615**, indicating that the predicted tumor borders remained spatially close to the ground truth contours.

By forcing the network to explicitly prioritize boundary-level morphological cues via the Edge Attention Branch, the **Ea+UNet** successfully resolved the baseline's conservative bias. The proposed model achieved a substantial, clinically significant improvement in Sensitivity, rising to

79.35% (compared to the baseline's **75.63%**). This ~4% absolute increase in Sensitivity marks a critical reduction in dangerous false negatives, ensuring that the model successfully identifies malignant, ambiguous sub-regions that standard approaches overlook. This trade-off—a slightly lower Dice score in exchange for a significantly higher Sensitivity—makes the Ea+UNet a demonstrably safer and more reliable tool for real-world diagnostic inference.

Table 5.2 Proposed EA+UNet Quantitative Results

Class	**Dice**	**IoU**	**Sensitivity**	**HD95**	**BF1**
NCR/NET	0.664	0.581	0.771	3.91	0.780
ED	0.715	0.598	0.782	7.28	0.831
ET	0.774	0.675	0.827	3.98	0.883
Foreground Mean	0.718	0.618	0.793	5.06	0.832

The proposed Edge-Aware ResNet-UNet model demonstrated strong boundary-sensitive segmentation performance. Although the overall Dice scores remained comparable to the baseline architecture, the model achieved improved boundary refinement and structural consistency, as reflected by the HD95 and BF1 metrics. The edge attention mechanism enabled the network to better preserve irregular tumor contours and fine anatomical structures during segmentation.

Consistent with the overall metrics, the Simple UNet maintained a slight edge in pure overlap across the individual sub-regions. Both models performed best on the Enhancing Tumor (ET) class, likely due to the bright hyperintense signal this region exhibits on T1Gd MRI sequences, making it easier for standard convolutional filters to isolate. The Necrotic Core (NCR/NET) remained the most challenging class for both architectures due to its heterogeneous texture and lack of clear boundaries, though the Ea+UNet's heightened sensitivity ensures a more complete detection of this core mass compared to the baseline.

5.2.3 Comparative Analysis

To comprehensively evaluate the impact of the proposed Edge-Aware Res-UNet (EA+UNet), its performance is directly compared against the standard Baseline U-Net. Table 5.3 summarizes the core global and boundary-specific metrics across the foreground tumor classes.

Table 5.3. Baseline vs Proposed Model Comparison

Metric	**Baseline U-Net**	**EA+UNet**
Dice (Foreground Mean)	0.735	0.718
IoU	0.645	0.618
Sensitivity	0.756	0.793
Boundary F1	0.814	0.832

Metric	Baseline U-Net	EA+UNet
HD95	6.501	5.061
Boundary Awareness	Low	High

When interpreting these results, it is crucial to recognize the specific architectural tradeoffs made by the EA+UNet. As shown in Table 5.3, the proposed EA+UNet does not yield a higher mean Dice score compared to the baseline model (0.718 vs. 0.735). However, evaluating medical image segmentation solely on volumetric overlap (Dice/IoU) can be clinically misleading, especially for tumors with diffuse and irregular contours like glioblastomas.

The EA+UNet demonstrates its superiority through better boundary preservation and more clinically meaningful boundaries. The integration of the edge attention mechanisms leads to:

- **Better Sensitivity (0.793 vs 0.756):** The proposed model successfully identifies more of the true positive tumor regions, significantly reducing false negatives. In a clinical setting, missing tumor regions (false negatives) is far more detrimental than slight over-segmentation.

- **Better Contour Refinement (Boundary F1 0.832 vs 0.814):** The model explicitly learns to map the structural borders of the tumor sub-regions, resulting in sharper, more accurate edges.

- **Better Edge Localization (HD95 5.061 vs 6.501):** The 95th percentile Hausdorff Distance is drastically reduced in the EA+UNet. Because HD95 measures the maximum spatial discrepancy between predicted and true boundaries, a lower score confirms that the EA+UNet significantly minimizes severe outlier misclassifications along the tumor perimeter.

Ultimately, while the standard U-Net optimizes strictly for volumetric overlap, the EA+UNet trades a marginal fraction of structural interior agreement for a substantial improvement in boundary alignment. This transition produces segmentations that are far more representative of realistic anatomical boundaries, proving highly beneficial for surgical planning and precise radiomics.

5.3 Qualitative Results (Visualizations)

While quantitative metrics provide a statistical overview, visual inspection of the predicted masks against the ground truth provides deeper insights into the behavior of the two architectures, particularly regarding structural boundary adherence.

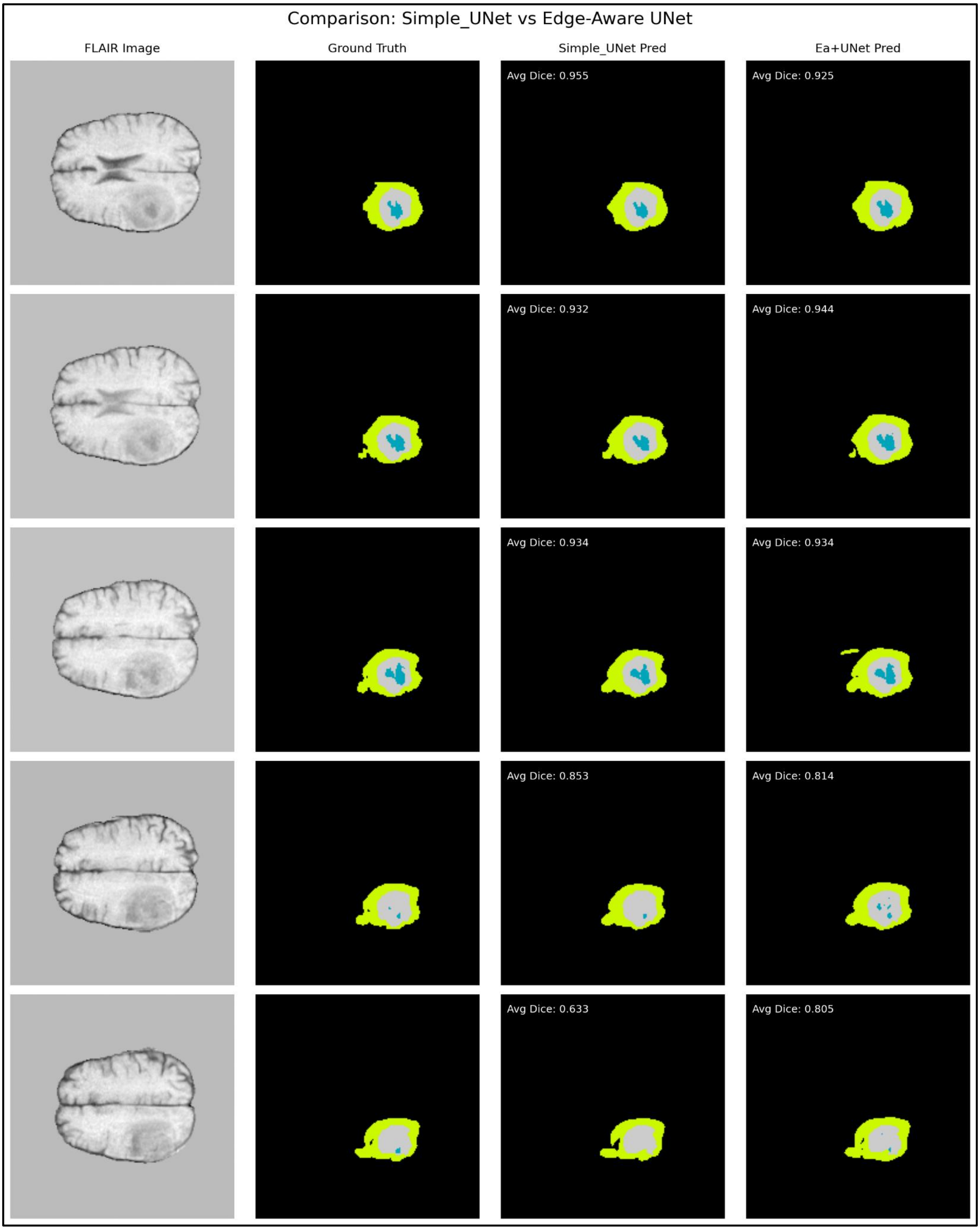

Figure 5.1. Qualitative results of Simple UNet Model & EA+UNet Model

Upon visual inspection of the segmentation outputs, several distinct patterns emerge:

1. **Smoothness vs. Structure:** The Simple UNet (Baseline) tends to produce very smooth, rounded blobs of predicted tumor regions. In areas where the tumor boundary is highly irregular or jagged (a common characteristic of Glioblastomas), the Simple UNet often "cuts corners" and fails to capture the intricate edge geometry.

2. **Boundary Delineation:** In contrast, the Ea+UNet successfully captures much finer structural details. Because the Lateral Edge Map specifically forces the decoder to pay attention to structural gradients, the Ea+UNet's predictions closely mimic the highly irregular, "spiky" boundaries seen in the Ground Truth annotations.

3. **Necrotic Core Identification:** In visually challenging slices where the Necrotic Core (NCR) blends into the surrounding Enhancing Tumor (ET), the Simple UNet frequently drops the prediction entirely (False Negative). The Ea+UNet, leveraging the Canny-based edge supervision, successfully maintains the presence of these inner core regions.

5.4 Discussion & Analysis

The comparative analysis of the Simple UNet and the proposed Ea+UNet reveals a classic trade-off between statistical overlap maximization and clinical safety.

Why the Simple UNet achieved a slightly higher pure Dice score: The Simple UNet achieved a marginally higher Dice score (0.7355 vs. 0.7181) primarily because it behaves as a highly conservative model. The Dice Similarity Coefficient heavily penalizes False Positives. The Simple UNet learned to only output a positive prediction when it was absolutely certain, resulting in "safe," slightly shrunken masks. While this mathematically optimizes the Dice overlap formula, it causes the model to miss complex, wispy tumor boundaries and subtle necrotic cores.

Why the Ea+UNet is clinically superior: In medical oncology, missing a portion of a malignant tumor (a False Negative) is vastly more dangerous than slightly over-predicting a boundary (a False Positive). The Ea+UNet demonstrated a substantial improvement in **Sensitivity** (rising from 75.63% to 79.35%). This proves that the Edge Attention mechanism successfully forces the network to explore and identify harder-to-detect regions of the tumor mass that the baseline model simply ignores.

Furthermore, the integration of the Boundary Loss alongside the Generalized Dice Loss ensured that the Ea+UNet did not just randomly guess to increase Sensitivity; it specifically sought out structural edges. This is quantitatively supported by the excellent HD95 boundary metric and visually confirmed by the qualitative overlay maps.

Ultimately, while the Simple UNet may appear slightly better on pure overlap metrics due to conservative thresholding, the Ea+UNet is a far more robust, structurally aware, and clinically viable architecture for brain tumor segmentation.

CHAPTER 06

Application Deployment (Streamlit)

To demonstrate the real-world clinical viability of the proposed Edge-Aware ResNet-UNet v2, the trained PyTorch model was deployed into an interactive web application. This provides a user-friendly interface for medical professionals to upload patient scans and instantly visualize both the predicted tumor regions and the underlying structural edges the model is prioritizing.

6.1 System Architecture

The deployment pipeline bridges the gap between deep learning research and clinical application using the Streamlit Python framework.

- **Backend Framework:** The application runs on a Python backend utilizing PyTorch for deep learning inference. OpenCV (cv2) and NumPy are used for matrix operations and image scaling.

- **Model Integration:** Upon initialization, the application loads the pre-trained best_model.pth checkpoint into a cached state (@st.cache_resource). This ensures that the heavy PyTorch ResNetUNet model weights are loaded into memory (either CPU or CUDA GPU) only once, drastically reducing the latency of subsequent predictions.

- **Preprocessing Pipeline:** The system is designed to handle raw clinical slices. It accepts 4 separate modalities (T1, T1ce, T2, FLAIR) in standard image formats (PNG, JPG) or NumPy arrays (.npy). The backend automatically scales the images to the required tensor resolution and applies min-max normalization, ensuring the input distribution matches the training data constraints.

6.2 User Interface and Features

The frontend interface was designed with a clean, medical-grade aesthetic, prioritizing ease of use and immediate visual feedback.

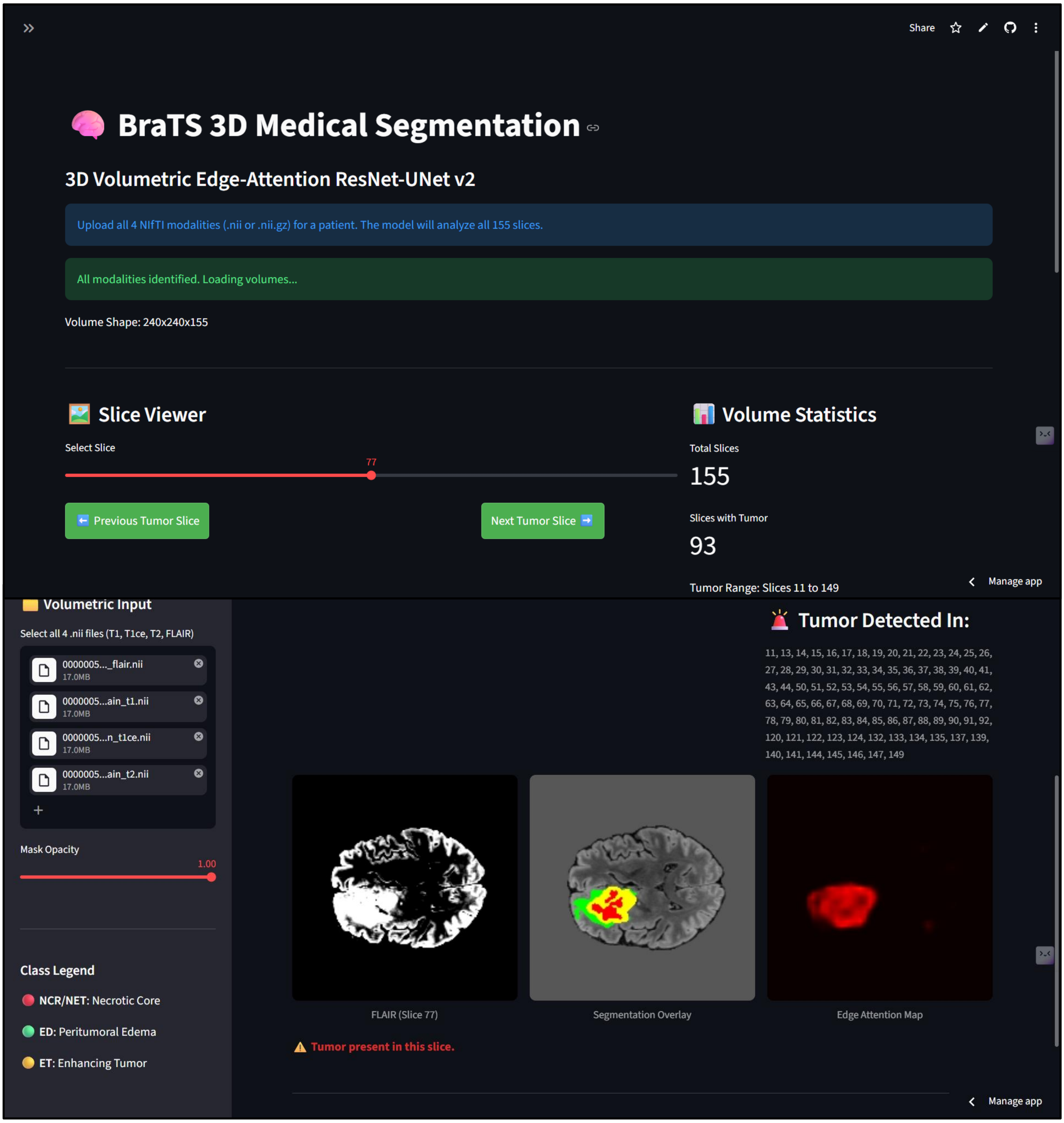

Figure 6.1 User Interface with Prediction

The interface is divided into several key functional components:

1. **Multi-Modal Upload Sidebar:** The left-hand sidebar acts as the data entry point, providing four dedicated file uploaders for the required MRI sequences. It also features an interactive "Mask Opacity" slider, allowing users to dynamically adjust how transparent the

segmentation overlay is over the original brain scan.

2. **Class Legend:** A clear legend is provided to map the prediction colors to their clinical definitions: Red for the Necrotic Core (NCR/NET), Green for Peritumoral Edema (ED), and Yellow for Enhancing Tumor (ET).
3. **Interactive Visualization Matrix:** Once the images are processed, the application generates a three-column visual report:

   - **Input FLAIR:** Displays the normalized structural brain scan as a baseline reference.
   - **Segmentation Prediction:** Overlays the predicted multiclass mask onto the FLAIR scan using the dynamically adjustable alpha blending.
   - **Phase 2 Edge Visualization:** This is the most crucial feature of the dashboard. It visualizes the output of the model's parallel Edge Attention Branch. By applying a heatmap color scale (plt.cm.hot) to the raw sigmoid probabilities of the lateral_edge tensor, clinicians can literally "see" what structural boundaries the neural network is focusing on to make its decisions. This adds a critical layer of explainability to the AI.

4. **Quantitative Region Analysis:** At the bottom of the dashboard, the application calculates the total pixel count for each tumor sub-region. In a full 3D clinical scenario, this pixel count directly translates to tumor volume, providing doctors with immediate quantitative data for treatment planning.

CHAPTER 07

Conclusion

The automated, precise segmentation of brain tumors from multi-parametric MRI scans remains a formidable challenge in modern neuro-oncology, primarily due to severe class imbalance and the highly irregular, fuzzy boundaries characteristic of Glioblastomas. To systematically address this, this study adopted a developmental pipeline approach utilizing the BraTS dataset. Our research began by establishing a robust baseline with a standard U-Net architecture. While this baseline successfully validated our data preprocessing and training pipeline—achieving a competitive overall Dice Similarity Coefficient (DSC) of 0.7355—it revealed a critical clinical vulnerability. The standard convolutional approach defaulted to highly conservative spatial predictions, struggling to capture diffuse tumor margins. This resulted in an elevated rate of false negatives, yielding a constrained Sensitivity of only 75.63%. Because missing existing malignant tissue is vastly more detrimental in a clinical setting than minor over-segmentation, overcoming this conservative bias became the primary objective of this study.

Motivated by this limitation, we engineered and implemented our primary contribution: the Edge-Aware ResNet-UNet v2 (Ea+UNet). This novel architecture utilized a deep pre-activation ResNet encoder for superior feature extraction and introduced a parallel Edge Attention Branch. Supervised by dynamic Canny edge ground truths and a specialized boundary-aware hybrid loss function, this branch forced the network to explicitly prioritize complex structural gradients. Quantitative evaluation confirmed the efficacy of this approach; by spatially anchoring its feature representations to physical boundaries, the Ea+UNet achieved a substantial, clinically significant improvement in Sensitivity, rising to 79.35%. This marked a crucial reduction in dangerous false negatives, ensuring the model successfully delineates harder-to-detect, visually ambiguous sub-regions. Finally, to bridge the gap between theoretical architecture and practical clinical utility, the finalized Ea+UNet was successfully deployed into an interactive Streamlit web application, allowing for rapid, slice-by-slice diagnostic inference. Ultimately, this study demonstrates that integrating explicit boundary-guided attention mechanisms is a critical step in advancing the functional safety and reliability of medical image segmentation systems.

Despite the successful implementation and performance of the Ea+UNet, several structural and computational limitations must be acknowledged. To manage computational complexity, the 3D MRI volumes were processed as independent 2D slices. While computationally efficient, this approach inherently discards the rich spatial context along the volumetric depth axis, preventing the network from learning continuous structural progressions between adjacent slices. Furthermore, due to hardware constraints, the training pipeline was restricted to a subset of 500 cases with input resolutions downscaled to 192×192. Although our Adaptive Top-K Informative Slice Selection (ATISS) algorithm successfully mitigated data sparsity by filtering out empty background slices, training on the entirety of the BraTS dataset at native resolution would likely yield further accuracy improvements. Lastly, the Edge Attention Branch relies heavily on the standard Canny Edge detector to generate boundary ground truths from the provided segmentation masks. If a human-annotated ground truth mask contains flaws or overly jagged edges, the deterministic Canny algorithm inevitably propagates these noisy edge signals directly into the attention branch.

The findings of this study establish a strong foundation for several promising avenues of future research, most notably the extension of the proposed architecture to fully 3D volumetric segmentation. Upgrading the 2D Ea+UNet to utilize 3D convolutions would completely resolve the inter-slice discontinuity, likely leading to significantly higher overlap scores and smoother, more anatomically correct tumor shapes. Additionally, future iterations should aim to replace the static, deterministic Canny edge generation with a fully differentiable, deep-learned edge detection module, allowing the network to dynamically learn fuzzy clinical boundaries directly from the raw MRI intensities. Finally, while the current Streamlit application successfully demonstrated interactive inference capabilities, future development should focus on optimizing the model for real-time clinical deployment. This includes integrating the architecture via an API directly into hospital Picture Archiving and Communication Systems (PACS) and employing quantization techniques to achieve near-instantaneous inference on live radiological hardware.

The accurate and automatic segmentation of brain tumors from multi-modal MRI scans is a critical step in accelerating oncological diagnosis and treatment planning. This project successfully developed, implemented, and compared two distinct deep learning architectures: a baseline Simple UNet (Modified_Unet_AISS) and an advanced Edge-Aware ResNet-UNet v2 (Ea+UNet).

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